

Statistics Practice Set — S21

S21 increases the difficulty slightly. The focus is now **choosing between measures when statistics conflict, interpreting skewness carefully, and making analyst-style business decisions.**

No basic formula-recall questions and no Population/Sample identification.

Q1 — Mean vs Median + Skewness

A company records the following monthly customer spending:

Spending
₹700
₹750
₹800
₹850
₹900
₹950
₹1,000
₹6,500

Answer:

1. Calculate the mean.
2. Calculate the median.
3. Which measure better represents typical customer spending?
4. Is the distribution likely right-skewed or left-skewed?
5. Explain why the mean and median are so different.
6. Does the difference between mean and median **prove** that ₹6,500 is an outlier? Explain.

Q2 — SD vs CV: Manufacturing

Two factories have the following production data:

Factory	Mean Production	SD
A	800 units	64
B	3,200 units	192

Answer:

1. Calculate CV for both factories.
2. Which factory has greater absolute variability?
3. Which factory has greater relative variability?
4. Which factory is more consistent relative to production?
5. Explain why SD alone is not enough for this comparison.

Q3 — Range vs IQR

Two stores record daily order values.

Store A

500, 520, 540, 550, 560, 580, 600

Store B

500, 520, 540, 550, 560, 580, 2,000

Answer:

1. Calculate the range for both.
2. Calculate Q1 for both.
3. Calculate Q3 for both.
4. Calculate IQR for both.

5. Which measure gives a more reliable comparison of the **middle 50%**?
 6. Explain why Store B's range changes dramatically while its IQR does not.
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Q4 — Five Number Summary: Skewness

A dataset has:

Statistic	Value
Minimum	10
Q1	18
Median	24
Q3	32
Maximum	75

Answer:

1. Calculate the lower whisker length.
 2. Calculate the upper whisker length.
 3. Which side has the longer tail?
 4. Identify the likely skewness.
 5. Give **two numerical reasons** for your conclusion.
 6. Can you conclude that 75 is definitely an outlier from this information alone? Explain.
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Q5 — Box Plot Business Comparison

Two departments have salary distributions:

Statistic	Dept A	Dept B
Minimum	₹22,000	₹15,000
Q1	₹30,000	₹25,000
Median	₹42,000	₹38,000
Q3	₹50,000	₹65,000
Maximum	₹58,000	₹1,10,000

Answer:

1. Which department has the higher median?
 2. Calculate the IQR for both.
 3. Calculate the range for both.
 4. Which department has greater variability?
 5. Which department is more consistent?
 6. Which department has the longer upper whisker?
 7. Which department shows stronger evidence of a long right tail?
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Q6 — Outlier Detection + Business Decision

Delivery times are:

12, 14, 15, 17, 18, 20, 21, 23, 26, 55

Using the IQR method:

1. Calculate Q1.
2. Calculate Q3.
3. Calculate IQR.
4. Calculate the lower fence.

5. Calculate the upper fence.
 6. Identify any outlier.
 7. Should the company use the mean or median to report typical delivery time?
 8. Explain your answer without saying merely “because there is an outlier.”
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Q7 — CV Ranking + Return Decision

Three investments have:

Investment	Mean Return	SD
A	10%	2%
B	18%	3.6%
C	28%	4.2%

Answer:

1. Calculate CV for A, B and C.
 2. Rank them from **lowest CV → highest CV**.
 3. Which investment has the highest return?
 4. Which investment has the lowest relative variability?
 5. Does the highest return also have the lowest CV?
 6. If an investor wants **high return with relatively low variability**, which option would you investigate first?
 7. Explain why CV is more useful here than SD alone.
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Q8 — Mean vs Median: Don't Overclaim

Two platforms report:

Platform	Mean Order Value	Median Order Value
A	₹1,500	₹1,000
B	₹1,200	₹1,100

Answer:

1. Which platform has the higher mean?
 2. Which platform has the higher median?
 3. Which platform shows stronger evidence of right-skewness?
 4. Which platform appears to have a larger gap between typical order value and average order value?
 5. Does Platform A's statistics **prove** that it contains an outlier?
 6. What additional method would you use to investigate possible outliers?
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Q9 — Choose the Best Statistical Tool

Choose from:

Mean, Median, SD, IQR, CV, Box Plot, Range

A.

A company wants the typical income of employees, but a few executives earn extremely high salaries.

B.

Two factories have very different average production levels, and management wants to compare relative consistency.

C.

A manager wants the spread of the middle 50% of delivery times.

D.

An analyst wants to visually compare six departments and identify possible outliers.

E.

Two machines have the same average output, and management wants to compare their absolute variation.

F.

Management wants the simplest measure of total spread from minimum to maximum.

G.

A dataset is approximately symmetric and has no major extreme values, and management wants the arithmetic average.

Q10 — Full Data Analyst Case

Two e-commerce platforms report:

Platform	Mean Order Value	Median	SD
A	₹1,800	₹1,200	₹360
B	₹2,200	₹2,000	₹330

Answer:

1. Calculate CV for both platforms.
2. Which platform has greater absolute variability?
3. Which platform has greater relative variability?
4. Which platform shows stronger evidence of right-skewness?
5. Which platform has the higher typical order value?
6. Which platform is more consistent relative to its mean?
7. If management wants **higher typical order value**, which platform is better?
8. If management wants **lower relative variability**, which platform is better?
9. Does Platform A's mean–median gap prove that it contains an outlier?
10. Give a final analyst conclusion using **mean + median + SD + CV together**.